ForgeGIS Studio™

An Operator Surface That Stays on One Host

ForgeGIS Studio™ is the browser workspace for the Forge suite: a live map, a pipeline canvas, a batch runner, and a natural-language agent acting on the operator's own map. Its deployment model is a single operator on a single machine, and that model is defended by default — the server binds to loopback, every path a caller supplies is sandboxed, resources are capped, and the agent reaches the map through its own authenticated door.

127.0.0.1

loopback by default

rebind it outward and Studio warns on every boot while it is off loopback

Sandboxed

every caller-supplied path

uploads confined and extension-checked; opted-in roots canonicalised, deny-listed and session-scoped

Separate

credential for the agent

the agent acts on the map through a token-validated bridge, not the browser's cookie

Why single-host is a decision

Studio's browser API carries no login because it is not meant to be reached by anyone but the operator at the machine. The prerequisites for any hosted deployment — real authentication, CSRF protection, rate limiting, TLS, an audit log and a sandbox re-review — are named in its design, not implied, so scaling outward is a planned step rather than a shortcut.

Bounded by construction

Upload and pixel caps, a GPU-memory estimate checked before any operation is dispatched, a bounded executor pool that refuses rather than thrashes, and a per-operation timeout cap the worst case one workstation has to absorb. Sessions expire, a request addressed to any host but loopback is refused — closing DNS rebinding — as are writes from other sites and other local pages, and zip and KMZ uploads are vetted before extraction and capped on real bytes as they inflate.

An agent you can bound

ForgeMind acts on Studio's map only through the authenticated `studio-mcp` bridge, whose tools are pinned by a versioned contract, under an operator-only sandbox that cannot reach paths a browser session added. On ForgeMind's compiled path a request becomes a validated pipeline or a named refusal, and every named result the agent places is recorded in the catalog.

New in 1.2.6 — correctness and hardening. Raster batches go to the suite's compute broker when it is installed, and run on the workstation when it cannot take them. Requests to any host but loopback are refused; Tools-panel calls are validated exactly as the ForgeGIS MCP server validates them; elevations carry the data's own vertical units; and the viewshed is drawn from the engine's 255 visible / 0 hidden encoding, as `gdal_viewshed` writes it.

CONTINUED ON REVERSE →

WHAT PROGRAMS GET

Terrain and analysis workflows, composed and batched

Hillshade, slope, viewshed, least-cost paths and the rest of ForgeGIS's raster and vector operations become nodes on a canvas; a graph proven on one area runs across a folder of inputs in batch — raster batches on the suite's compute broker when one is installed, the rest on the workstation with each tile's GPU memory checked as it is dispatched — with a report at the end. Pipelines converted from QGIS models carry their import warnings, including the one case that runs but answers differently.

The operator's own data

Load from the ForgeData catalog — including a shared, curated catalog for the whole workstation — or drop KML, KMZ and zipped shapefiles straight onto the map. Area-windowed reads pull just the pixels you asked for, and a registered-but-missing dataset is caught when you pick it, not when it fails to read.

Answers stated honestly

Elevations print in the units the data states, and "unknown units" when nothing says; lengths and areas are measured on the WGS84 ellipsoid. A raster with no georeferencing is marked "position unknown" rather than drawn near Null Island, and a rotated GeoTIFF the reader cannot place is refused with its remedy. Over-size geometry is drawn from file if geographic and refused by name if not — never projected coordinates drawn as degrees.

Checked against independent references

Release 1.2.6 came out of a testing and validation audit, and its fixes are pinned by tests shown failing first. An end-to-end suite drives Studio in a headless browser against the real engine and checks both the screen and the value under it against GDAL; measurements are pinned to GeographicLib and pyproj values.

> **Footprint.** An opt-in component of the Forge suite, installed and started by the Seaglass Control Center™ on one Windows x64 workstation, on port 8081 behind loopback. The suite brings its own Java runtime; the agent bridge is a standalone executable. The basemap and URL layers need a network; rasters render offline.

Request the Technical Brief and an evaluation build

The Technical Brief covers the architecture, the security posture in full, the batch runner, the agent bridge and deployment. ForgeGIS Studio is part of the Forge suite and is not sold separately. The Forge suite is commercially licensed; evaluation builds are available to qualified programs.

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